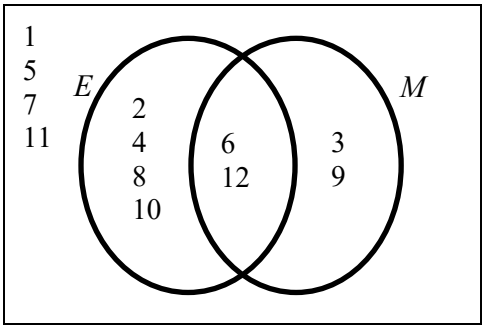
| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 6(c) | $\tan b = \frac{8}{4}$ oe | **M1** | |
| | 63.43… | **A1** | |
| 6(d) | Yes correct reason | **2** | **B1** for evidence that all three triangles each have angles 63.4°, 26.6° and 90°<br>**B1** for yes and statement that triangles are similar because they have the same 3 angles oe |
| 7(a)(i) | $4x-9$ cao | **3** | **B2** for $x+(x-3)+2(x-3)$ oe<br>or **B1** for $k(x-3)$ seen $k=1, 2$ or $3$ oe |
| 7(a)(ii)(a) | 11 nfww | **3** | **M1** for *their* **(a)(i)** = 35<br>**M1** for rearranging *their* $(ax+b)=35$<br>to $ax=35-b$ or $x+\frac{b}{a}=\frac{35}{a}$ or better |
| 7(a)(ii)(b) | 5 | **2** | **FT** *their x*<br>**M1** for (*their x* $-3)\times 2$ soi<br>or **B1** for [Pierre makes] 16 |
| 7(b)(i) | Half-past two shown correctly on clock face | **1** | |
| 7(b)(ii) | 105 | **2** | **M1** for $\frac{3.5}{12}[\times 360]$ oe |
| 7(c) | $8.64 \times 10^5$ | **2** | **M1** for $60\times 60\times 24\times 10$<br>or **B1** for figs 864<br>If 0 scored, **SC1** for correctly changing *their answer* into standard form provided *their answer* $> 10\,000$ |
| 7(d) | 18 02 | **2** | **M1** for $3\frac{1}{2}\times 8$<br>or **B1** for 1736 seen |
| 7(e) | 16 cao | **4** | **B1** for LCM=24 soi<br>**M1** for $\frac{365}{24}$ or $\frac{364}{24}$ or $\frac{365}{48}$ or $\frac{364}{48}$<br>**A1** for 15 or 15.2<br>or 15.16 to 15.17 or 15.20 to 15.21<br>If **A0** scored, **SC1** for 7.60[4..] or 7.58[3..] and 8 final answer |
| 8(a)(i) | Angle [in a] semicircle | **1** | |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 8(a)(ii) | 30 | **2** | **M1** for $\frac{6\times10}{2}$ |
| 8(a)(iii) | 11.7 or 11.66… | **2** | **M1** for $[x^2 =]\ 6^2+10^2$ or better |
| 8(b) | $[r =]\sqrt{\frac{A}{\pi}}$ | **2** | **M1** for $\frac{A}{\pi}=r^2$ or $\sqrt{A}=\sqrt{\pi}\times r$ |
| 8(c) | 17.4 or 17.37 to 17.38… | **4** | **M3** for $81-\pi\left(\frac{\sqrt{81}}{2}\right)^2$ oe<br>OR<br>**M1** for $\sqrt{81}$<br>and **M1** for $\pi\left(\frac{their\sqrt{81}}{2}\right)^2$<br>and **M1** for $81-their\pi\left(\frac{\sqrt{81}}{2}\right)^2$ |
| 9(a)(i) |  | **2** | **B1** for 2 or 3 regions correct |
| 9(a)(ii) | 8 | **1** | **FT** *their* **(a)(i)** |
| 9(a)(iii) | $\frac{1}{6}$ oe | **2** | **FT** *their* **(a)(i)**<br>**B1FT** for 2 or *their* $\mathrm{n}(E\cap M)$ |
| 9(b) | No 2 is even and a prime oe | **1** | |